

Partial Result: Amenable-Radical Obstruction for Reduced L^p Group Banach Algebras

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Status

This is a partial result for the reverse-implication question in Phillips, *Simplicity of reduced group Banach algebras*, arXiv:1909.11278. Phillips asks whether simplicity of $F_r^p(G)$ implies the Powers property. The result below does not prove that implication, but shows that simplicity already rules out non-trivial amenable normal subgroups.

Source Question

Phillips proves that $C_r^*(G)$ simple implies that $F_r^p(G)$ and $B_r^{p,*}(G)$ have the Powers property and hence are simple. In the introduction, around source lines 408–414, he writes that the reverse implications are not addressed and that it is not known whether simplicity of $F_r^p(G)$ implies the Powers property.

Idea of the Proof

Let N be an amenable normal subgroup of G . The quotient representation on $\ell^p(G/N)$ can be approximated by the regular representation on $\ell^p(G)$: spread a vector on G/N uniformly over a finite Folner set in each coset. For a fixed finitely supported element of $\mathbb{C}[G]$, only finitely many translations inside N are relevant, and amenability makes the Folner set almost invariant under all of them. This proves that the quotient map on group rings is contractive for the reduced L^p norms.

Theorem

Theorem 1. *Let G be a discrete group, let N be an amenable normal subgroup, and let $q : G \rightarrow G/N$ be the quotient map. Let $1 \leq p < \infty$. Then the induced homomorphism*

$$\pi : \mathbb{C}[G] \rightarrow \mathbb{C}[G/N], \quad \pi(g) = q(g),$$

is contractive for the reduced L^p operator norms. Hence it extends to a contractive homomorphism with dense range

$$\Pi_N : F_r^p(G) \longrightarrow F_r^p(G/N).$$

Proof. Write $\lambda_{G,p}$ and $\lambda_{G/N,p}$ for the left regular representations on $\ell^p(G)$ and $\ell^p(G/N)$. It suffices to prove that

$$\|\lambda_{G/N,p}(\pi(a))\| \leq \|\lambda_{G,p}(a)\| \quad (a \in \mathbb{C}[G]).$$

Fix $a = \sum_{g \in E} a_g g \in \mathbb{C}[G]$ with E finite. Choose a section $\sigma : G/N \rightarrow G$ of q . For $g \in E$ and $t \in G/N$, define

$$\alpha(g, t) = \sigma(q(g)t)^{-1} g \sigma(t) \in N.$$

First let $\xi \in \ell^p(G/N)$ have finite support L . For a finite non-empty set $F \subset N$, define an isometry $V_F : \ell^p(G/N) \rightarrow \ell^p(G)$ by

$$(V_F \xi)(\sigma(t)n) = |F|^{-1/p} \xi(t) \mathbf{1}_F(n) \quad (t \in G/N, n \in N).$$

For a basis vector δ_t and $g \in E$, the vectors $\lambda_{G,p}(g) V_F \delta_t$ and $V_F \lambda_{G/N,p}(q(g)) \delta_t$ are both supported in the coset $\sigma(q(g)t)N$, and their difference has norm

$$\left(\frac{|\alpha(g, t) F \triangle F|}{|F|} \right)^{1/p}.$$

Since N is amenable and the set of all $\alpha(g, t)$ with $g \in E$ and $t \in L$ is finite, there is a net of finite non-empty sets F such that all these quantities tend to zero. Therefore

$$\|\lambda_{G,p}(a) V_F \xi - V_F \lambda_{G/N,p}(\pi(a)) \xi\| \longrightarrow 0.$$

Using that V_F is an isometry, we get

$$\begin{aligned} \|\lambda_{G/N,p}(\pi(a)) \xi\| &= \lim_F \|V_F \lambda_{G/N,p}(\pi(a)) \xi\| \\ &\leq \limsup_F \|\lambda_{G,p}(a) V_F \xi\| \leq \|\lambda_{G,p}(a)\| \|\xi\|. \end{aligned}$$

Finite-support vectors are dense in $\ell^p(G/N)$, so the desired operator norm inequality follows. The extension to completions is then immediate, and its range contains $\mathbb{C}[G/N]$, hence is dense in $F_r^p(G/N)$. $\square$

Corollary 1. *Let $1 < p < \infty$. If G has a non-trivial amenable normal subgroup, then $F_r^p(G)$ is not simple. The same is true for $B_r^{p,*}(G)$.*

Proof. Let $1 \neq n \in N$. The map Π_N sends the canonical generator w_n to 1, so $w_n - 1 \in \ker \Pi_N$. This element is nonzero in $F_r^p(G)$: the canonical trace τ_p satisfies $\tau_p(w_n) = 0$ and $\tau_p(1) = 1$. The kernel is proper because $\Pi_N(1) = 1$. Thus $F_r^p(G)$ has a nonzero proper closed ideal.

For $B_r^{p,*}(G)$, the norm on $\mathbb{C}[G]$ is the maximum of the reduced p -norm of a and the reduced p -norm of a^* . Since $\pi(a^*) = \pi(a)^*$, Theorem 1 gives the same contractive dense-range homomorphism, and the same kernel argument applies. $\square$

Corollary 2. *If $1 < p < \infty$ and either $F_r^p(G)$ or $B_r^{p,*}(G)$ is simple, then the amenable radical of G is trivial.*

Limitations

This is only a necessary condition. It does not prove that simplicity of $F_r^p(G)$ implies the Powers property, nor that it implies $C_r^*(G)$ is simple. Possible remaining counterexamples must have trivial amenable radical; the dynamical obstructions to C^* -simplicity are not addressed here.

Relation to Gardella–Thiel

Gardella–Thiel, arXiv:1408.6137, proved a quotient theorem for $F_\lambda^p(G) \rightarrow F_\lambda^p(G/N)$ when N is amenable and G/N is finite, and noted that the finite quotient assumption was likely unnecessary. The Folner-spreading proof above gives the contractive dense-range map without that finite quotient assumption, which is enough for the simplicity obstruction used here.

References

- N. Christopher Phillips, *Simplicity of reduced group Banach algebras*, arXiv:1909.11278v1.
- E. Gardella and H. Thiel, *Functoriality of group algebras acting on L^p -spaces*, arXiv:1408.6137.